Smart Home Sensor — Quick Reference

Full build instructions: scan the QR code

Web flasher (on a computer, in a browser with Web Serial: Chromium, Chrome, Edge, Opera):
`alexanderkutschera.com/smart_home_sensor`

1. Wire the BME680

Solder from the **back** of the sensor board, so the wires leave it behind the components.

BME680	VCC	GND	SDA	SCL
Board	**3V3** (not 5 V)	GND	**GPIO3**	**GPIO2**

Swapped SDA/SCL is the most common reason a board reports no sensor.

2. Flash

Web flasher → select the workshop image → **Connect** → port `USB JTAG/serial debug unit` → **Install**. Needs a USB-C **data** cable. No port listed? Unplug, hold **BOOT**, plug in, release after ~2 s.

3. Set up on the device

First boot opens the Wi-Fi network **`SHS-xxxxxxxx-Setup`** (no password) → browse to **`192.168.4.1`** → **Configure WiFi** (2.4 GHz only) → **Setup**: give the device a name → *Live readings & connection test* → **Send a test reading** (green ✓ **201** or **200** = stored) → **Finish**.

Back into setup later: press RESET twice quickly.

4. Your readings

`diy-sensor.org/dashboard/device/<device-id>` — the device ID is on the display and on the setup page. It comes from the chip and never changes. New readings arrive every **5 minutes**; the display itself refreshes every few seconds.

5. Air quality

IAQ accuracy: 0 stabilizing · **1–2** calibrating (hours) · **3** trusted. A drop back to 1 is normal, BSEC is rebuilding its baseline. **IAQ:** 0–50 good · 51–100 moderate · 101–150 light · 151–200 moderate · 201–300 heavy · 300+ severe pollution. **Temperature** reads high (the board warms the sensor). After 20–30 min, *add* the remaining error to the offset in the setup page: `new = current + (reported − real)`. It starts at 5 °C, not 0.

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